

EFPB Simulation Results

Every result produced so far: the SUMO and surrogate experiments, the SUMO pilot, calibration and the demand study

Runs of 19 September 2026 on one laptop. Companion to EFPB_Experiment_Run_Report.pdf (what was done and how).

1. Read this first

- **These are traffic-simulation results only.** Nothing here tests explanations, comprehension, trust or any other part of the HCI study; that part of the research plan has not been carried out.
- **Status: exploratory.** The matrix experiments (section 3) were started without confirmation that the preregistration draft had been registered first. Unless it was registered before 17:39 on 2026-09-19, treat them as exploratory, not confirmatory.
- **Two backends.** SUMO/TraCI is the real traffic simulator (vehicles and signals in SUMO; *pedestrians are a virtual queue* because the network has no validated walking areas). The surrogate is a fast in-process approximation used for engineering validation; its numbers should not be reported as traffic evidence.
- **Descriptive statistics only.** Means over cells (cells weighted equally) and paired differences on identical scenarios and seeds, with standard errors. The preregistered models are not implemented, so there are no significance tests.
- **Objectives.** *Person delay* is the occupancy-weighted mean wait of everyone who arrived after warm-up (people still waiting at the end count with the wait so far). *Max wait / limit* is the largest wait divided by its 100 s limit; above 1 the limit is exceeded. Lower is better for both.

Result sets in this document

Result set	Backend	Runs	Seeds	Status
Matrix: 9 nominal + 4 stress cells x 4 controllers	SUMO	520	4001–4010 (10)	Exploratory; 30-seed version kept for future work
Matrix: confirmatory_core, robustness, stress	Surrogate	1920 (1880 analysed)	1001–1030, 2001–2010, 1101–1130	Exploratory; engineering validation
Pilot: 3 scenarios x 4 controllers	SUMO	36	9001–9003 (3)	Time-boxed pilot
Calibration selection	Surrogate / SUMO	3200 / 2800	301–305	Parameter selection, calibration data only
Calibration confirmation and delta sensitivity	Surrogate / SUMO	2,400 / 700	321–350 / 361–370	Fresh seeds; parameters frozen
Demand (saturation) study	Both	160 / 45	301–305 / 301–303	Chose demand levels S and O

2. Headline results

- **Nominal demand, SUMO (9 cells).** efpb has the lowest mean person delay: 6.31 s, against 6.45 for efficiency, 6.68 for equal_bargaining and 8.49 for the tuned fixed-time plan. It is lower than each of them in all 9 cells (paired differences -0.13 ± 0.02 , -0.36 ± 0.04 , -2.18 ± 0.16 s). Its max wait is lower than efficiency's (-0.05 ± 0.01) and about equal to equal_bargaining's (-0.00 ± 0.01).
- **The tuned fixed-time plan has the lowest max wait** in the nominal cells (0.27 against 0.41 for efpb in SUMO) because every phase comes round within its 37 s cycle, but at a person delay 2.2 s higher. Neither objective alone decides a winner.

- **The surrogate agrees for nominal demand:** efpb ties efficiency on person delay ($+0.00 \pm 0.02$ s) with a lower max wait (-0.19 ± 0.01); fixed-time again has the lowest max wait.
- **Under saturation there is no clear ordering.** In SUMO efpb, efficiency and fixed-time are within about 7 s on person delay with large standard errors, and the over-saturated cell T_OO fails for every controller. In the surrogate efficiency has the lowest person delay under stress.
- **equal_bargaining collapses under saturation** (SUMO stress: person delay 339 s, max wait 18 times the limit) because it almost never serves pedestrians while vehicles are queued. This is how the baseline is implemented; whether that is the intended baseline is undecided.
- **Safety:** no collisions and no emergency stops in any of the 520 SUMO matrix runs; after adding yellow and all-red clearance the SUMO pilot also had none (79 emergency stops without it).
- **Calibration:** a demand-based fixed-time plan with a 37 s cycle wins in both backends; in the surrogate only efpb changes from its defaults (kappa 4, eta 2, service duration 15 s), in SUMO every controller keeps its defaults. The delta tolerance barely matters.

3. Matrix experiments

Frozen parameters, fresh paired seeds. SUMO: 13 demand cells x 4 controllers x 10 seeds, 4,200 s per run (600 s warm-up, 3,600 s evaluation). Surrogate: confirmatory_core (9 cells, 30 seeds), robustness (8 of 9 scenarios analysed, 10 seeds) and stress (4 cells, 30 seeds). Cell T_XY is pedestrian level X, vehicle level Y (L, M, H = light, medium, high; S = near saturation, O = over saturation).

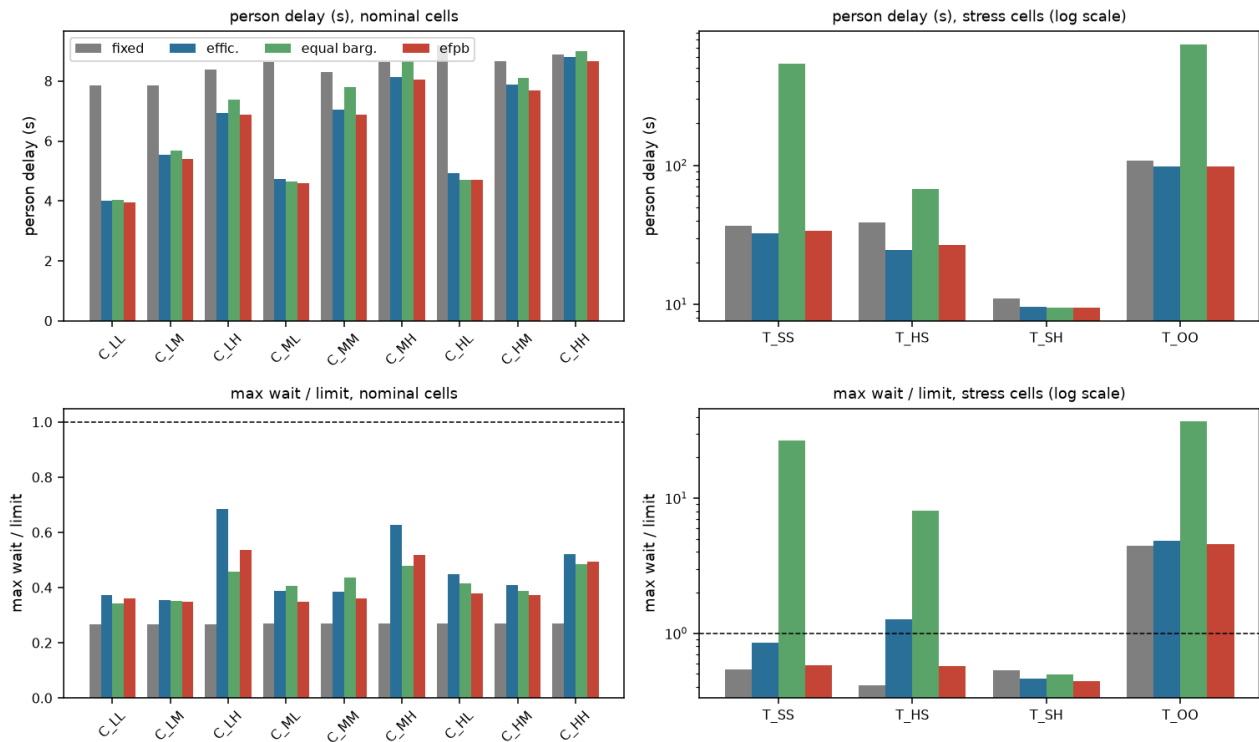
3.1 SUMO/TraCI (virtual pedestrians)

Means over cells (nominal: 9 cells, stress: 4 cells).

Group	Controller	Person delay (s)	Max wait / limit	Ped wait (s)	Veh wait or stop (s)	Unservd ped / unfinished veh
nominal	fixed_time	8.5	0.27	11.2	7.7	0.4 / 4.8
nominal	efficiency	6.4	0.47	8.7	6.2	0.4 / 4.1
nominal	equal_bargaining	6.7	0.42	6.0	6.8	0.3 / 4.6
nominal	efpb	6.3	0.41	8.7	6.0	0.3 / 4.1
stress	fixed_time	48.6	1.48	72.3	37.6	32.0 / 41.1
stress	efficiency	41.1	1.85	34.6	129.1	8.1 / 38.3
stress	equal_bargaining	338.9	18.04	807.5	324.7	424.2 / 36.1
stress	efpb	42.0	1.54	23.5	133.8	5.2 / 39.6

efpb against each controller, paired on scenario and seed: mean difference $\pm$ standard error (cells where efpb is lower / cells). Negative = efpb lower, i.e. better.

Group	efpb minus	Person delay (s)	Max wait / limit
nominal	fixed_time	-2.18 ± 0.16 (9/9)	$+0.14 \pm 0.01$ (0/9)
nominal	efficiency	-0.13 ± 0.02 (9/9)	-0.05 ± 0.01 (9/9)
nominal	equal_bargaining	-0.36 ± 0.04 (9/9)	-0.00 ± 0.01 (5/9)
stress	fixed_time	-6.63 ± 9.99 (4/4)	$+0.06 \pm 0.97$ (1/4)
stress	efficiency	$+0.90 \pm 13.91$ (1/4)	-0.31 ± 1.42 (4/4)
stress	equal_bargaining	-296.93 ± 79.45 (4/4)	-16.50 ± 2.37 (4/4)



Per cell (lowest value in each block in bold; demand levels of pedestrians / vehicles):

Cell	Demand ped/veh	delay fixed	delay effic.	delay equal barg.	delay efpb	wait fixed	wait effic.	wait equal barg.	wait efpb
C_LL	L / L	7.8	4.0	4.0	4.0	0.27	0.38	0.34	0.36
C_LM	L / M	7.9	5.5	5.7	5.4	0.27	0.35	0.35	0.35
C_LH	L / H	8.4	6.9	7.4	6.9	0.27	0.69	0.46	0.54
C_ML	M / L	8.6	4.7	4.7	4.6	0.27	0.39	0.41	0.35
C_MM	M / M	8.3	7.1	7.8	6.9	0.27	0.39	0.44	0.36
C_MH	M / H	8.6	8.1	8.7	8.1	0.27	0.63	0.48	0.52
C_HL	H / L	9.2	4.9	4.7	4.7	0.27	0.45	0.42	0.38
C_HM	H / M	8.7	7.9	8.1	7.7	0.27	0.41	0.39	0.38
C_HH	H / H	8.9	8.8	9.0	8.7	0.27	0.52	0.49	0.49

Cell	Demand ped/veh	delay fixed	delay effic.	delay equal barg.	delay efpb	wait fixed	wait effic.	wait equal barg.	wait efpb
T_SS	S / S	36.5	32.5	537.4	33.6	0.54	0.85	26.76	0.58
T_HS	H / S	38.6	24.7	67.7	26.8	0.41	1.26	8.06	0.57
T_SH	S / H	11.0	9.5	9.5	9.4	0.54	0.46	0.49	0.45
T_OO	O / O	108.3	97.5	740.9	98.0	4.45	4.85	36.85	4.58

3.2 Surrogate (engineering validation only)

Means over cells (nominal: 9 cells, robustness: 8 cells, stress: 4 cells).

Group	Controller	Person delay (s)	Max wait / limit	Ped wait (s)	Veh wait or stop (s)	Starvation violations
nominal	fixed_time	12.3	0.30	10.9	12.7	0.00
nominal	efficiency	9.9	0.70	12.3	10.0	0.17

Group	Controller	Person delay (s)	Max wait / limit	Ped wait (s)	Veh wait or stop (s)	Starvation violations
nominal	equal_bargaining	10.1	0.57	8.9	10.5	0.00
nominal	efpb	9.9	0.52	13.2	9.6	0.00
robustness	fixed_time	12.7	0.33	11.4	13.0	0.00
robustness	efficiency	11.0	0.59	12.4	10.8	0.01
robustness	equal_bargaining	11.4	0.61	10.9	11.5	0.27
robustness	efpb	11.2	0.50	14.2	10.6	0.00
stress	fixed_time	44.5	1.55	72.9	36.7	575.06
stress	efficiency	36.1	1.55	29.4	38.2	290.04
stress	equal_bargaining	67.1	6.09	235.7	23.2	361.59
stress	efpb	45.6	1.87	34.3	48.9	396.49

efpb against each controller, paired on scenario and seed: mean difference $\pm$ standard error (cells where efpb is lower / cells). Negative = efpb lower, i.e. better.

Group	efpb minus	Person delay (s)	Max wait / limit
nominal	fixed_time	-2.42 $\pm$ 0.08 (9/9)	+0.21 $\pm$ 0.00 (0/9)
nominal	efficiency	+0.00 $\pm$ 0.02 (5/9)	-0.19 $\pm$ 0.01 (9/9)
nominal	equal_bargaining	-0.19 $\pm$ 0.02 (8/9)	-0.05 $\pm$ 0.01 (8/9)
robustness	fixed_time	-1.47 $\pm$ 0.18 (5/8)	+0.17 $\pm$ 0.01 (0/8)
robustness	efficiency	+0.18 $\pm$ 0.05 (2/8)	-0.09 $\pm$ 0.01 (7/8)
robustness	equal_bargaining	-0.20 $\pm$ 0.04 (6/8)	-0.11 $\pm$ 0.02 (7/8)
stress	fixed_time	+1.13 $\pm$ 0.75 (2/4)	+0.32 $\pm$ 0.08 (2/4)
stress	efficiency	+9.56 $\pm$ 1.47 (0/4)	+0.32 $\pm$ 0.09 (3/4)
stress	equal_bargaining	-21.45 $\pm$ 5.24 (3/4)	-4.22 $\pm$ 0.76 (4/4)

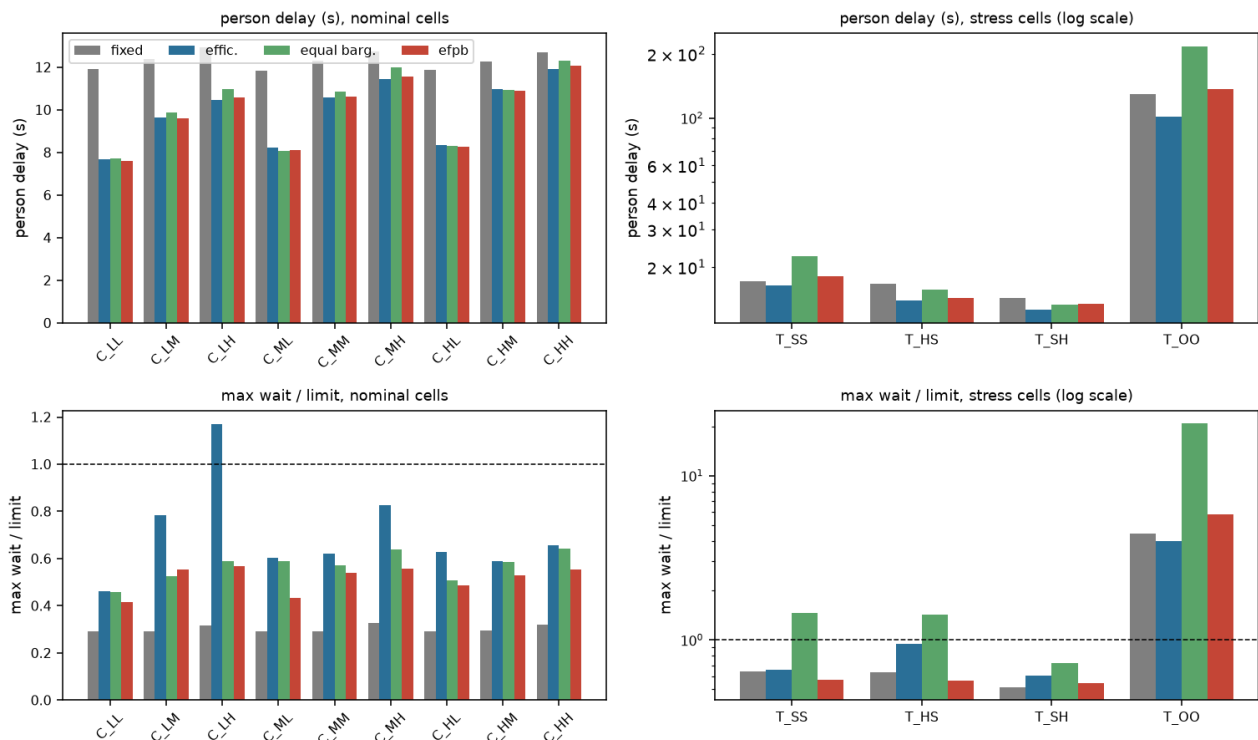


Figure: person delay and max wait per cell for the four controllers (stress cells on a log scale; the dashed line is the wait limit).

Per cell (lowest value in each block in bold; demand levels of pedestrians / vehicles):

Cell	Demand ped/veh	delay fixed	delay effic.	delay equal barg.	delay efpb	wait fixed	wait effic.	wait equal barg.	wait efpb
C_LL	L / L	11.9	7.7	7.7	7.6	0.29	0.46	0.46	0.41
C_LM	L / M	12.4	9.7	9.9	9.6	0.29	0.78	0.53	0.55
C_LH	L / H	13.0	10.5	11.0	10.6	0.32	1.17	0.59	0.57
C_ML	M / L	11.9	8.2	8.1	8.1	0.29	0.60	0.59	0.43
C_MM	M / M	12.3	10.6	10.9	10.6	0.29	0.62	0.57	0.54
C_MH	M / H	12.8	11.4	12.0	11.6	0.33	0.83	0.64	0.56
C_HL	H / L	11.9	8.3	8.3	8.3	0.29	0.63	0.51	0.48
C_HM	H / M	12.3	11.0	10.9	10.9	0.29	0.59	0.59	0.53
C_HH	H / H	12.7	11.9	12.3	12.1	0.32	0.66	0.64	0.55

Cell	Demand ped/veh	delay fixed	delay effic.	delay equal barg.	delay efpb	wait fixed	wait effic.	wait equal barg.	wait efpb
T_SS	S / S	17.1	16.3	22.3	18.1	0.65	0.66	1.47	0.57
T_HS	H / S	16.6	13.9	15.6	14.3	0.64	0.94	1.42	0.57
T_SH	S / H	14.2	12.6	13.3	13.4	0.52	0.60	0.72	0.54
T_OO	O / O	130.1	101.5	217.1	136.7	4.42	3.99	20.77	5.81

Robustness scenario	delay fixed	delay effic.	delay equal barg.	delay efpb	wait fixed	wait effic.	wait equal barg.	wait efpb
S10_low_demand	11.9	7.5	7.6	7.4	0.29	0.52	0.51	0.42
S12_sensor_error	12.3	12.3	12.3	12.3	0.30	0.30	0.30	0.30
S4_tail_wait	12.4	9.7	10.0	9.4	0.29	0.73	0.56	0.54
S5_ns_wait	12.3	10.5	10.7	10.5	0.30	0.65	0.58	0.54
S6_conflict	12.9	12.0	12.4	12.1	0.33	0.65	0.66	0.55
S7_emergency	12.3	10.5	10.7	10.5	0.30	0.65	0.58	0.54
S8_repeated_imbalance	14.1	13.3	14.1	14.2	0.47	0.57	0.77	0.56
S9_saturation	13.3	12.8	13.6	13.4	0.40	0.64	0.93	0.55

Excluded from every mean and comparison: S11_accessibility (controller marks every vehicle plan unsafe for the whole run; vehicles are never served). S12 (sensor error) gives identical numbers for every controller because they all fall back to the same fixed plan, and S7 (emergency) is a single-second request with no visible effect; both are valid but uninformative.

4. SUMO pilot (time-boxed)

36 runs: 3 scenarios (balanced, pedestrian-heavy, vehicle-heavy) x 4 controllers x 3 seeds (9001–9003, never used for calibration); 900 s evaluation. fixed_time uses the tuned demand-based 37 s plan. Means of 3 seeds; no collisions and no emergency stops.

Scenario	Controller	Veh/h	Stop time (s)	Wait p95 (s)	Queue	Ped served	Ped wait (s)	Switches	Clear. %
S1_balanced	fixed_time	792	7.5	18.7	1.63	65.0	10.4	73	18.7
S1_balanced	efficiency	783	7.3	16.7	1.55	65.3	9.0	72	18.1
S1_balanced	equal_bargaining	788	8.3	20.3	1.76	63.3	6.1	72	17.0
S1_balanced	efpb	780	6.9	17.0	1.47	65.3	8.7	72	18.1
S2_pedestrian_heavy	fixed_time	792	7.5	18.7	1.63	101.3	11.6	73	18.7
S2_pedestrian_heavy	efficiency	789	8.5	17.3	1.81	99.7	8.5	73	17.6

Scenario	Controller	Veh/h	Stop time (s)	Wait p95 (s)	Queue	Ped served	Ped wait (s)	Switches	Clear. %
S2_pedestrian_heavy	equal_bargaining	780	9.0	17.7	1.91	99.0	6.7	74	16.7
S2_pedestrian_heavy	efpb	789	8.1	17.0	1.72	99.7	8.1	73	17.7
S3_vehicle_heavy	fixed_time	1239	9.4	19.0	3.02	65.0	10.4	73	18.7
S3_vehicle_heavy	efficiency	1219	7.4	17.7	2.34	64.7	13.2	72	20.0
S3_vehicle_heavy	equal_bargaining	1236	9.0	22.7	2.83	64.7	6.5	73	17.9
S3_vehicle_heavy	efpb	1229	7.4	18.7	2.31	63.3	10.6	71	20.1

- **Effect of yellow/all-red clearance** (pilot with the static 90 s fixed-time plan so timing is not mixed in): mean vehicle stop time without / with clearance – fixed_time 16.1 / 16.5 s; efficiency 4.6 / 7.7 s; equal_bargaining 5.5 / 8.8 s; efpb 4.4 / 7.5 s. Emergency stops: 79 without, 0 with.
- **Effect of tuning the fixed-time plan** (same seeds): mean vehicle stop time 16.5 s with the static 90 s plan against 8.1 s with the tuned demand-based 37 s plan.
- In the pilot no controller is clearly ahead on vehicles (stop times within about 1 s); equal_bargaining has the lowest pedestrian waits at these light demand levels. Three seeds only.

5. Calibration results

Every controller was tuned under the same budget (at most 30 configurations from its own search space, default always included) on calibration scenarios and seeds 301–305. Each controller's pick is the configuration closest to the ideal point (lowest delay and lowest max wait, both scaled 0–1 within that controller); ties keep the default. The pick was recorded before any confirmation run, then compared with the default on fresh seeds and adopted unless worse in both objectives.

5.1 surrogate

Controller	Configs tried	Delay range (s)	Max wait range	Default (delay / max wait)	Pick	Adopted?
fixed_time	14	18.6–38.2	0.50–1.46	18.6 / 0.50	default	yes
efficiency	27	17.1–17.8	0.70–0.86	17.1 / 0.73	default	yes
equal_bargaining	9	21.2–23.7	0.99–1.55	21.2 / 0.99	default	yes
efpb	30	17.6–23.2	0.70–1.27	18.2 / 0.87	eta 2, kappa 4, delta 0.05, horizon 60, duration 15	yes

Configuration confirmed on fresh seeds	Controller	Person delay (s)	Max wait / limit	Worst scenario
horizon 60, switch cost 5, duration 25 (default)	efficiency	16.4	0.68	1.43
eta 1.0, kappa 2.0, delta 0.05, horizon 60, duration 25 (default)	efpb	17.2	0.73	2.21
eta 2, kappa 4, delta 0.05, horizon 60, duration 15	efpb	17.4	0.69	1.92
efpb delta 0	efpb	17.4	0.69	1.88
efpb delta 0.02	efpb	17.4	0.69	1.88
efpb delta 0.1	efpb	17.3	0.69	1.91
efpb delta 0.2	efpb	17.3	0.69	1.93
horizon 60, duration 25 (default)	equal_bargaining	19.5	0.99	3.59
cycle 37, plan demand_based (default)	fixed_time	18.2	0.51	1.52

Frozen (configs/frozen_surrogate.yaml): {"signal": {"fixed_cycle_s": 37, "fixed_timing": "demand_based"}, "per_controller": {"efpb": {"controller": {"debt_eta": 2, "debt_kappa": 4}, "signal": {"service_duration_s": 15}}}}

delta	Person delay (s)	Max wait / limit	Max wait vs picked
0	17.38	0.6865	-0.0050 ± 0.0045
0.02	17.38	0.6865	-0.0050 ± 0.0045
0.05	17.37	0.6915	+0.0000 ± 0.0000
0.1	17.33	0.6874	-0.0041 ± 0.0022
0.2	17.34	0.6863	-0.0053 ± 0.0028

5.2 SUMO

Controller	Config s tried	Delay range (s)	Max wait range	Default (delay / max wait)	Pick	Adopted?
fixed_time	14	20.4–46.7	0.50–1.75	20.4 / 0.50	default	yes
efficiency	27	16.9–17.4	0.49–0.53	16.9 / 0.50	horizon 30, switch cost 0, duration 25	no (kept default)
equal_bargaining	9	30.3–44.4	2.16–2.42	30.3 / 2.16	default	yes
efpb	30	16.0–39.1	0.35–1.36	16.8 / 0.42	eta 2, kappa 2, delta 0.05, horizon 90, duration 35	no (kept default)

Configuration confirmed on fresh seeds	Controller	Person delay (s)	Max wait / limit	Worst scenario
horizon 30, switch cost 0, duration 25	efficiency	16.7	0.51	0.86
horizon 60, switch cost 5, duration 25 (default)	efficiency	16.6	0.51	0.88
eta 1.0, kappa 2.0, delta 0.05, horizon 60, duration 25 (default)	efpb	17.3	0.44	0.58
eta 2, kappa 2, delta 0.05, horizon 90, duration 35	efpb	19.3	0.48	1.46
efpb delta 0	efpb	19.3	0.48	1.46
efpb delta 0.02	efpb	19.3	0.48	1.46
efpb delta 0.1	efpb	19.3	0.48	1.46
efpb delta 0.2	efpb	19.3	0.48	1.46
horizon 60, duration 25 (default)	equal_bargaining	30.6	2.23	8.99
cycle 37, plan demand_based (default)	fixed_time	24.1	0.62	2.52

Frozen (configs/frozen_sumo.yaml): {"signal": {"fixed_cycle_s": 37, "fixed_timing": "demand_based"}}

delta	Person delay (s)	Max wait / limit	Max wait vs picked
0	19.29	0.4846	+0.0000 ± 0.0000
0.02	19.29	0.4846	+0.0000 ± 0.0000
0.05	19.29	0.4846	+0.0000 ± 0.0000
0.1	19.29	0.4830	-0.0016 ± 0.0015
0.2	19.33	0.4826	-0.0020 ± 0.0023

5.3 Fixed-time cycle length under stress

Cycle (s)	Surrogate: delay / max wait / worst scenario	SUMO: delay / max wait / worst scenario
37	18.6 / 0.50 / 1.53	20.4 / 0.50 / 1.59
40	20.9 / 0.60 / 1.92	21.3 / 0.64 / 2.27
45	21.7 / 0.62 / 1.95	20.8 / 0.58 / 1.81
50	19.8 / 0.54 / 1.30	20.8 / 0.53 / 1.34
60	22.5 / 0.62 / 1.44	24.1 / 0.58 / 1.28
75	24.6 / 0.67 / 1.26	27.5 / 0.63 / 1.12
90	27.1 / 0.74 / 1.21	31.4 / 0.72 / 1.09
120	34.3 / 0.95 / 1.36	39.3 / 0.94 / 1.15

Demand-based plans (splits from each scenario's nominal demand). Both backends pick 37 s, the shortest cycle that fits three 10 s minimum greens plus clearance. Longer cycles lower the worst-scenario wait but raise the overall delay. Static plans with proportional splits had a much higher max wait at every cycle length.

6. Demand study: where the system saturates

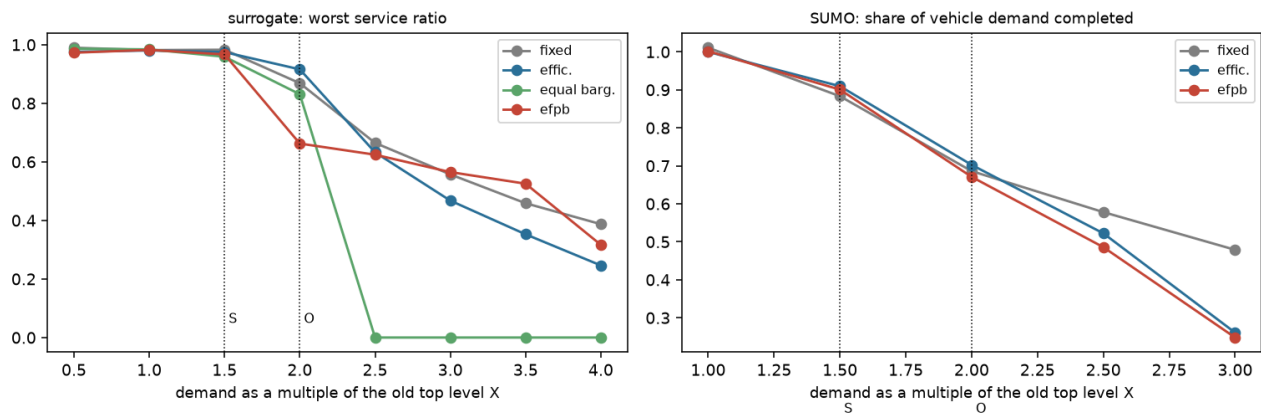


Figure: all controllers at their then-current defaults on a ladder of demand levels. The old top level X (0.15 pedestrians and 0.44 vehicles per second) is about half of capacity. Dotted lines mark the two levels added: S = 1.5x X (0.225 / 0.66 per second, near saturation) and O = 2.0x X (0.30 / 0.88, over saturation).

x X	ped / veh per s	surrogate fixed	surrogate effic.	surrogate equal barg.	surrogate efpb
0.5	0.075 / 0.22	0.99 / 0.29	0.98 / 0.54	0.99 / 0.56	0.97 / 0.48
1.0	0.150 / 0.44	0.98 / 0.38	0.98 / 0.53	0.98 / 0.53	0.98 / 0.53
1.5	0.225 / 0.66	0.98 / 0.53	0.97 / 0.57	0.96 / 1.30	0.97 / 0.55
2.0	0.300 / 0.88	0.87 / 1.53	0.92 / 1.80	0.83 / 3.54	0.66 / 3.33
2.5	0.375 / 1.10	0.66 / 2.96	0.63 / 6.43	0.00 / 8.99	0.62 / 6.49
3.0	0.450 / 1.32	0.56 / 4.07	0.47 / 8.99	0.00 / 8.98	0.56 / 7.51
3.5	0.525 / 1.54	0.46 / 5.08	0.35 / 8.99	0.00 / 9.00	0.53 / 7.49
4.0	0.600 / 1.76	0.39 / 5.65	0.25 / 8.99	0.00 / 8.99	0.32 / 8.69

Surrogate cells: worst service ratio / max wait per limit.

x X	veh per s	SUMO fixed	SUMO effic.	SUMO efpb
1.0	0.44	1.01 / 3.8	1.00 / 3.3	1.00 / 3.4
1.5	0.66	0.88 / 19.7	0.91 / 18.5	0.90 / 19.2
2.0	0.88	0.69 / 29.4	0.70 / 30.8	0.67 / 36.7
2.5	1.10	0.58 / 39.2	0.52 / 44.3	0.49 / 46.2
3.0	1.32	0.48 / 39.8	0.26 / 57.7	0.25 / 57.6

SUMO cells: share of vehicle demand completed / mean vehicles queued. SUMO's vehicle throughput plateaus near 2,200 vehicles per hour (about 0.62 per second).

7. What these results support, and what they do not

- **Supported (descriptively):** under nominal demand, efpb matches or beats the adaptive baselines on person delay and has a lower max wait than efficiency-only control; a properly tuned fixed-time plan is a strong baseline on tail waits. These patterns hold in both backends.
- **Not supported:** any claim that efpb is better under saturation (no clear ordering); any claim about explanations, comprehension, calibrated reliance or fairness perception; any inferential (significance) claim.
- **Sensitivity to the objectives.** Max wait favours short fixed cycles by construction, person delay favours adaptive control, and equal cell weights let the over-saturated cell dominate averages. Report both objectives and the per-cell tables.
- **Baselines.** equal_bargaining collapses under saturation as implemented; the ablation variants (no starvation guard, no efficiency filter, static weights, weighted sum) are not implemented, so no ablation results exist.
- **Scope.** One four-arm intersection; virtual pedestrians (no walking time, crossing or pedestrian clearance interval modelled); SUMO cells cover demand only, not emergency, sensor-error, accessibility or bursty scenarios.
- **Sample size.** The SUMO matrix has 10 seeds per cell where the research plan's design has 30; the 30-seed run is kept as future work (about 50 more minutes on this laptop). The standard errors treat cell-and-seed pairs as independent and ignore differences between cells, so they are not reliable for the 4-cell stress group.

8. Where the numbers come from

Result	Files (inside efpb-research-pipeline/)	How to reproduce
SUMO matrix	data/sumo_matrix/runs/ (runs), data/results/*_sumo.csv	python run_sumo_matrix.py; python analyze_matrix.py
Surrogate matrix	data/matrix/ (runs), data/results/*_surrogate.csv	logs/run_matrix_surrogate.sh; python analyze_matrix.py
SUMO pilot	data/sumo_runs/, data/processed/sumo_metrics.csv	logs/run_sumo_pilot.sh; python aggregate_sumo.py
Calibration	data/calibration_sweep/, data/sumo_calibration/ (selection.json, summary, freeze_report.md)	sweep_calibration.py / sweep_sumo_calibration.py, then freeze_calibration.py
Frozen parameters	configs/frozen_surrogate.yaml, configs/frozen_sumo.yaml	written by freeze_calibration.py
Demand study	data/saturation_study/	python saturation_study.py
This document	docs/make_results.py	python docs/make_results.py (needs reportlab, pyyaml, matplotlib)

Every stored run carries a simulator version and a configuration fingerprint, and the simulators refuse to reuse a run made with another configuration. The raw run folders are large and are not committed; the summary CSVs and reports listed above are.